Feature Importance & Business Insights

Santander Customer Satisfaction · Author: Hazel · Model: XGBoost (n=300, depth=5, lr=0.05) on 308 cleaned features with simple FE · Dataset: 76,020 customers, overall unsatisfied rate = 3.96%

1. Executive Summary

We analyzed feature importance using two complementary methods: **gain-based importance** (how often the model uses each feature) and **SHAP analysis** (how each feature shifts individual predictions). The two methods produced different rankings, and SHAP — generally considered more reliable for business interpretation — reveals a clearer story.

> ***Three factors drive ~50% of all prediction signal:*** *customer age (var15, 30.7%), current account balance (saldo_var30, 13.2%), and historical average balance (saldo_medio_var5_hace3, 5.9%).* ***High-risk customer profile:*** *middle-aged (30–60), low or zero balance, low historical activity.*

2. Top 5 Features by SHAP Importance

Rank	Feature	SHAP Share	Interpretation
1	var15	30.7%	Customer age (community consensus)
2	saldo_var30	13.2%	Current account balance
3	saldo_medio_var5_hace3	5.9%	Historical avg balance (3 periods ago)
4	saldo_medio_var5_ult3	3.6%	Recent avg balance (last 3 periods)
5	saldo_medio_var5_hace2	3.4%	Historical avg balance (2 periods ago)

Methodological note: Gain-based importance ranked *ind_var30* first, but SHAP ranks it 17th. Gain tends to overstate the importance of binary indicator variables that are used as frequent split points. SHAP measures actual prediction impact and is the more trustworthy ranking for business interpretation.

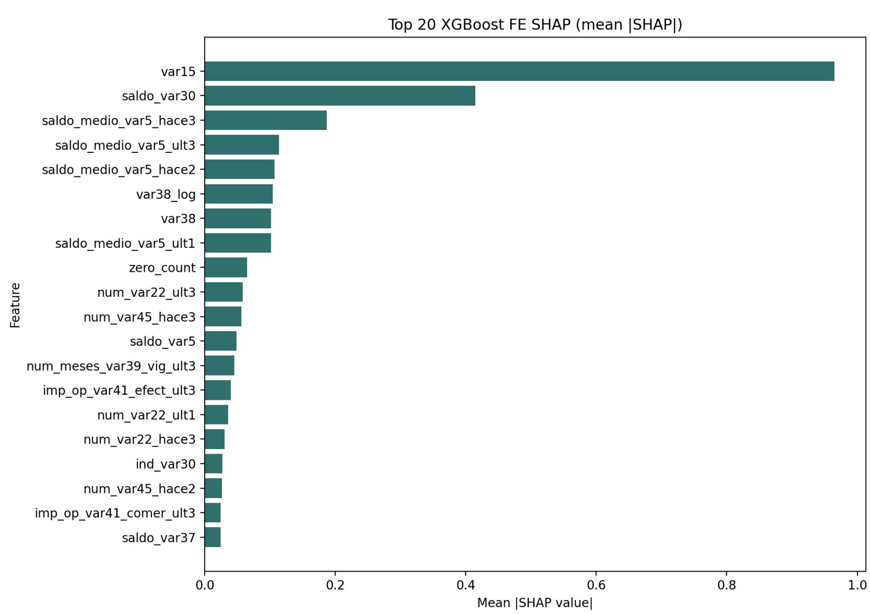

Figure 1. Top 20 features by mean |SHAP value|. var15 alone accounts for nearly a third of the model's prediction signal.

3. SHAP Directional Findings

The beeswarm plot below shows directional impact for the top 20 features. Each dot is one customer; color = feature value (red=high, blue=low); horizontal position = impact on the model's prediction (positive = pushes toward "unsatisfied").

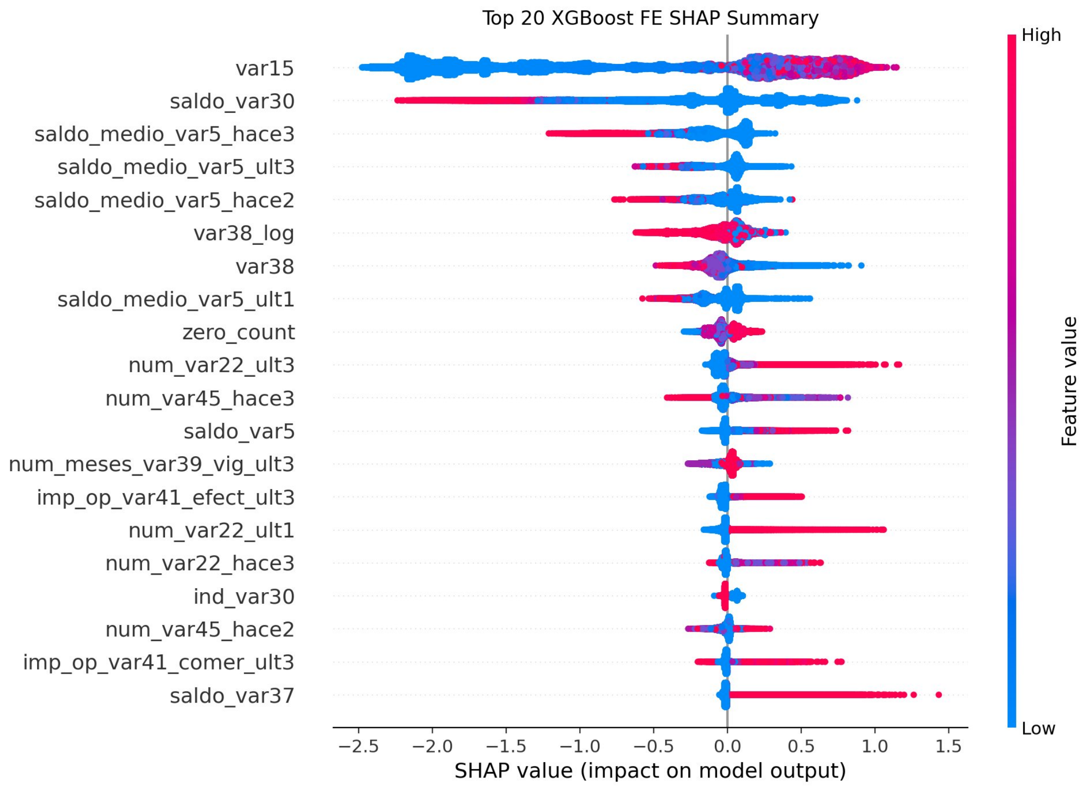

Figure 2. SHAP beeswarm summary plot.

Feature	Direction	Business meaning
var15	Low → strongly negative; high → mildly positive	Older = more dissatisfied (peaks 45–55)
saldo_var30	High → strongly negative	Higher balance = more satisfied
saldo_medio_var5_hace3	High → negative	Higher historical balance = more satisfied
var38	High → negative	Higher customer value = more satisfied
zero_count	High → mildly positive	Less data populated = slightly more dissatisfied

4. Business Insights from Interpretable Features

4.1 var15 (Age) — Middle-aged customers are most dissatisfied

Age bucket	Sample size	Unsatisfied rate	Lift vs overall
<23	1,212	0.00%	0.00x
23–30	40,650	1.65%	0.42x
30–45	20,968	6.80%	1.72x
45–60	**9,224**	**7.65%**	**1.93x**
60+	3,966	5.17%	1.31x

Key finding: dissatisfaction peaks in the 45–60 age group at nearly 2x the overall rate. SHAP dependence analysis (Figure 3) confirms a sharp increase in risk starting at age 23 and peaking around 45–55.

> ***Caveat on the <23 bucket:*** *the 1,212 customers here show 0% dissatisfaction. Inspection of raw data shows ages 5–22 are sparsely populated (a few dozen per year) before a sharp jump at age 23 (20,170 customers). These are likely sub-accounts, minors, or inactive accounts rather than primary customers, and should not be interpreted as "young customers are perfectly satisfied." The 23–30 bucket is the appropriate baseline for young primary customers, where dissatisfaction is still well below average (1.65%).*

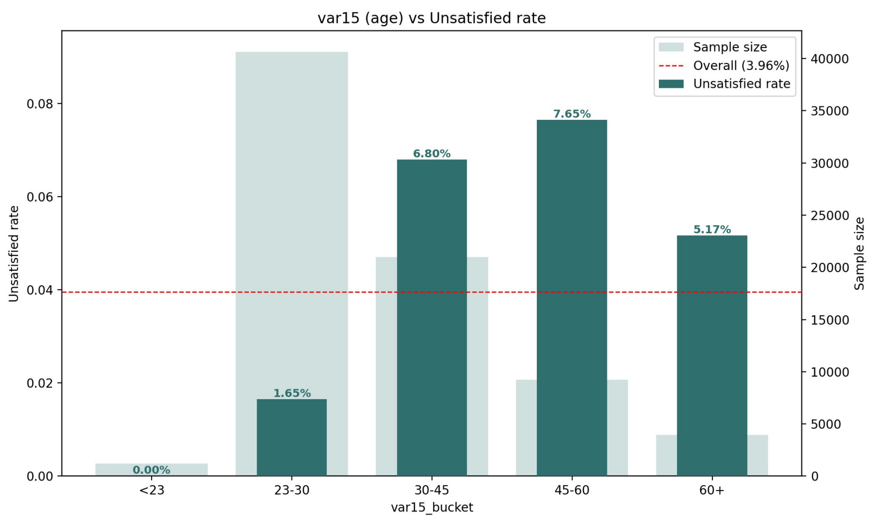

Figure 3. var15 (age) bucket vs unsatisfied rate. Dashed line = overall 3.96% baseline.

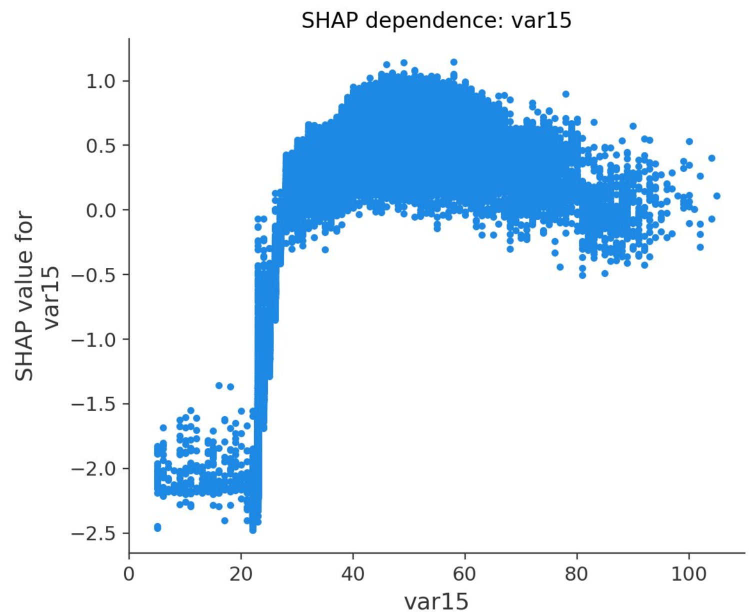

Figure 4. SHAP dependence plot for var15. Note the sharp transition at age ~23.

4.2 saldo_var30 (Balance) — Zero-balance drives most dissatisfaction

Balance	Sample size	Unsatisfied rate	Lift
≤ 0	**20,430**	**8.84%**	**2.24x**
> 0	55,590	2.16%	0.55x

Zero-balance customers are **4x more likely** to be unsatisfied than customers with positive balance. This corroborates Austin's earlier EDA finding that low-activity customers are the highest-risk segment.

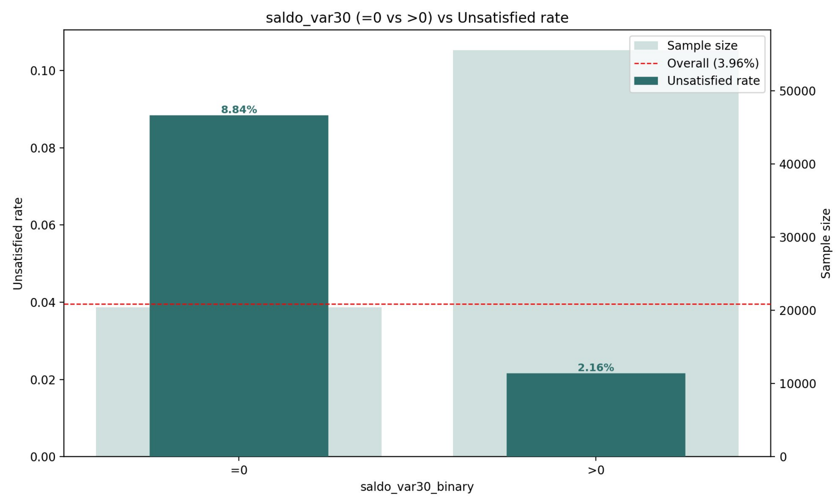

Figure 5. saldo_var30 (=0 vs >0) vs unsatisfied rate.

4.3 ind_var30 (Product holding indicator)

ind_var30	Sample size	Unsatisfied rate	Lift
0 (not held)	**20,310**	**8.79%**	**2.22x**
1 (held)	55,710	2.19%	0.55x

The 0 group of *ind_var30* (20,310) overlaps almost exactly with the zero-balance group of *saldo_var30* (20,430), suggesting these two features represent the same underlying product. Customers without this product form a clear churn-risk segment.

4.4 ind_var26_cte — Small but high-risk segment

ind_var26_cte	Sample size	Unsatisfied rate	Lift
0	73,925	3.88%	0.98x
1 (held)	**2,095**	**6.68%**	**1.69x**

Only 2.7% of customers hold this product, but their dissatisfaction rate is 1.7x the average. This small subgroup warrants a separate root-cause investigation by the bank.

5. Recommendations for Presentation

- **Lead with the three-factor narrative:** age + balance + activity. Concrete and actionable for a business audience.
- **Use SHAP rankings, not gain rankings,** when communicating which features matter most.
- **Flag the *ind_var26_cte* segment** as a small but high-risk group worth a targeted intervention.
- **Acknowledge the var15 < 23 caveat** when presenting age results to avoid misinterpretation.

6. Deliverable Files

File	Purpose
xgboost_fe_top20_shap_summary.png	SHAP beeswarm – main presentation visual
xgboost_fe_top20_shap_bar.png	Top 20 SHAP bar chart
xgboost_fe_shap_dependence_*.png (×6)	Per-feature SHAP dependence plots
xgboost_fe_top20_shap.csv	SHAP ranking table
business_var15_age.png	Age bucket vs unsatisfied rate
business_saldo_var30_binary.png / _quintile.png	Balance vs unsatisfied rate
business_ind_var30.png / business_ind_var26_cte.png	Product holding vs unsatisfied rate
business_top_features_summary.csv	Combined business analysis table